The enlarged direct comparison: what has been removed and what remains

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The count comparison now needs only an own-count moment estimate and an incoming-tail estimate. The same conditional third moments enter both the own-count loss and the marginal penalty; pricing them together avoids paying twice for incompatible worst cases. On the right, the first incoming count also retains the two different jump bounds. All later counts have one common variance bound, which gives a complete factorial tail. This is the selected route for further simplification.

Theorem 1 (Union of the reviewed component domains)**.** *Let $n \geq 42$, $Y = B + U$ with B Bernoulli of parameter p and $\mathbb{E}(U \mid B) = 0$. Put $v = p(1-p)$, $s = \mathbb{E}U^2 > 0$, $\lambda = ns$, $X = (Y-p)/\sqrt{v+s}$, $\rho = \mathbb{E}|X|^3$, and $\rho_B = (1-2v)/\sqrt{v}$. Assume $|H| \leq 46/25$. Let S_n be the sum of independent copies of X and F_B the $\mathrm{Bin}(n,p)$ CDF. Each row below supplies an integer k such that, with $\delta = k - np$,*

$$\begin{aligned}\Delta_C = \sqrt{n}\{\mathbb{P}(S_n \leq H) - \Phi(H/\sqrt{n})\} - C\rho \\ - \sqrt{n}\{F_B(k) - \Phi(\delta/\sqrt{nv})\} + C\rho_B < 0.\end{aligned}$$

The support convention in the table is $-A \leq U_0 \leq a_0$, $-a_1 \leq U_1 \leq A$, with $a_j \geq 0$.

p	A	$a_0 + a_1$	$0 < \lambda \leq$	C
$[1/4, 2/5]$	$7/10$	$17/250$	$1/100$	$[819/2000, 41/100]$
$[2/5, 53/100]$	$3/5$	$9/125$	$3/40$	$[2/5, \infty)$

The left row has the stronger conclusion $\Delta_C < -\lambda/100$. The right row asserts only strict negativity on its enlarged domain. The previously established right theorem with $\lambda \leq .03$ retains its stronger margin, and the older $\lambda \leq .02$ theorem for all $C \geq 0$ is unchanged.

Proof. In the left row choose the phase $u = np + H\sqrt{v+s} - k$ in $[-.218, .782]$ and apply LEFT-01 in `leftcount.tex`. In the right row choose $u \in [-.272, .728]$. Both intervals have width one, so such anchors exist. If $\lambda \leq .03$, the existing theorem in `counts-v11.tex` applies. For $.03 \leq \lambda \leq .075$, the negative inner phase $u \in [-.15, 0]$ is covered by `compound-proof.tex`; the other phases are covered by `outer.tex`. Endpoints are included in both adjoining intervals, and no phase or noise gap remains. Each component uses the same actual conditional laws, and each retains strict outgoing and closed incoming events. No optimizer of a moment relaxation is substituted for an iid law. □

How this joins to the acquired witness

For an original standardized witness Z, let B label its two conditional parts and $m_j = \mathbb{E}(Z \mid B = j)$, with separation $\ell = m_1 - m_0 > 0$. Then $m_0 = -p\ell$, $m_1 = (1-p)\ell$. Put

$$U = (Z - m_B)/\ell, \qquad Y = B + U = (Z - m_0)/\ell.$$

These are the actual marginal conditional variables. If $\theta = \mathbb{E}(Z - m_B)^2$, then

$$s = \theta/\ell^2, \qquad v + s = 1/\ell^2, \qquad (Y - p)/\sqrt{v + s} = Z, \qquad \lambda = n\theta/\ell^2.$$

Thus the raw sum threshold H and the original law are preserved. The v11 left support rays imply the left amplitude and inward assumptions; the v10/v11 right strips imply the right ones. Those acquisition results retain their contact, heat, actual-index and sharp-predecessor hypotheses. The direct comparison itself has none of those premises, but it cannot supply its own strip or threshold bounds.

At the sharp Bernoulli constant $C_E = (3 + \sqrt{10})/(6\sqrt{2\pi})$, both coefficient restrictions hold. The external strict finite-n Bernoulli theorem bounds the bracket at every integer anchor. Hence a witness of nonnegative excess satisfying either row is excluded. At zero noise the law is itself Bernoulli and that external theorem applies directly. The negative signed discrepancy still requires reflection and the appropriate attained or one-sided CDF witness; this corollary does not bypass those steps.

Exact dependency changes

Within the two rows, once their premises have been acquired, no additional half-amplitude confinement, harmonic or fractional-part gain, marked crossings, count-curvature estimate or count grid is needed to close the count comparison. Each replacement is explicit:

- The left own count and moment penalty are bounded together by Lemma 2 of `leftcount.tex`; its guarded endpoint prices produce an additional $7s/4$ reserve. Separately discarding the unfavorable left moment correction would be false.
- The right negative inner count uses physical multipliers $1 - \delta/(nq)$ and $1 + \delta/(np)$. Its first incoming probability is bounded by a centered two-jump Poisson hinge. A finite complement evaluates that hinge with its whole tail included. Its 291 closed rational boxes cover noise, phase and variance allocation; sample size is handled by concavity in $1/n$, not by a finite list of sample sizes.
- The other right phases use the bound $\operatorname{Var} W_j \le (5/2)\lambda$ for every count, followed by a factorial continuation. Two polynomials on one rectangle and five scalar endpoint comparisons complete those phases.

The left proof has 88 scalar checks. The right negative-inner acceptance has 23 normalization checks, 200 quartic cells and 91 Cantelli cells; each quartic cell has 256 endpoint choices and five Bernstein coefficients. The other right phases have 56 scalar checks and 144 coefficients. These are bounded executable inequalities, not a certificate-free proof. The external quadratic-hinge comparison remains an analytic dependency.

What is still missing

The acquired left range above $\lambda = .01$ and right range above $\lambda = .075$ remain open up to their retained panel-specific budgets. The direct geometry's global bound $\lambda < .616$ is much larger than either closed range. There is no global propagation theorem here. An independent finite base through 41 remains missing, and the retained 11,264-cell coarse Fourier entry has not been replaced. These prevent a claim of the full sharp conjecture.

No original theorem, certificate, manuscript or baseline source was removed to obtain this progress. The earlier exploratory common-allocation shortcut has an explicit admissible counterexample, as does the proposed convexity of the centered compound hinge in its allocation.

The new proofs use neither shortcut. Reviews, source hashes, failed controls and fresh run receipts are supplied separately; arithmetic success does not certify acquisition, attainment or the external comparison theorem.